Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: PH RANCH #2

Physical address of dairy:

Number and Street	City	County	Zip Code
2618 S Highway 59	Merced	Merced	95340

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1974

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X066-X100-X013-XXXX X066-X100-X014-XXXX

B. OPERATORS

Van Waremerdam, Paul

Operator name: Van Waremerdam, Paul

Telephone no.: (123) 000-0000 Landline (123) 000-0000 Cellular

Mailing Address Number and Street	City	State	Zip Code
6335 Oakdale RD	Winton	CA	95388

This operator is responsible for paying permit fees.

C. OWNERS

Veldhuis, Jeanette

Legal owner name: Veldhuis, Jeanette

Telephone no.: (123) 000-0000 Landline Cellular

Mailing Address Number and Street	City	State	Zip Code
6335 Oakdale RD	Winton	CA	95388

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	700	110	10	0	0	0
Maximum number	720	110	20	0	0	0
Average number	700	110	10	0	0	0
Avg live weight (lbs)	1,400	1,500	100	0		

Predominant milk cow breed: Holstein

Average milk production: 95 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 21,726.62 *tons per reporting period*

Total nitrogen from manure: 282,643.12 *lbs per reporting period*

After ammonia losses (30% loss applied): 197,850.18 *lbs per reporting period*

Total phosphorus from manure: 47,492.52 *lbs per reporting period*

Total potassium from manure: 145,553.96 *lbs per reporting period*

Total salt from manure: 354,889.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 38,592,000 *gallons*

Total nitrogen generated: 70,033.91 *lbs*

Total phosphorus generated: 13,631.14 *lbs*

Total potassium generated: 438,250.29 *lbs*

Total salt generated: 589,063.54 *lbs*

	38,592,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	38,592,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Ag Well Central	Ground water
Ag Well New Well	Ground water
Ag Well North	Ground water
Canal Water	Surface water
Domestic Well Barn	Ground water

Source Description	Type
Domestic Well House J's	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
#1 (VL 3)	100	100	2	process wastewater	X066-X100-X014-XXXX
#2 (VL 1)	60	60	2	process wastewater	X066-X100-X013-XXXX
Totals for areas that were used for application	160	160	4		
Totals for areas that were not used for application					
Land application area totals	160	160	4		

B. CROPS AND HARVESTS

#1 (VL 3)

Field name: #1 (VL 3)

10/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 100 Plant date: 10/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	1,000.00 *ton*	Dry-weight		18.1	25,000.00	2,150.00	22,300.00		18.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.00	409.50	35.22	365.27	2,964.78

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 100 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/22/2023	2,943.00 *ton*	Dry-weight		67.3	13,800.00	2,380.00	9,150.00		5.93

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.43	265.61	45.81	176.11	1,141.36

#2 (VL 1)

#2 (VL 1)

Field name: #2 (VL 1)

10/12/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 60 Plant date: 10/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	600.00 *ton*	Dry-weight		73.0	31,400.00	2,220.00	21,300.00		17.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.00	169.56	11.99	115.02	955.80

06/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 60 Plant date: 06/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/23/2023	1,454.60 *ton*	Dry-weight		67.3	13,800.00	2,380.00	9,150.00		5.93

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	24.24	218.80	37.74	145.07	940.21

NUTRIENT BUDGET

A. LAND APPLICATIONS

#1 (VL 3) - 10/10/2022: Oats, silage-soft dough

Field name: #1 (VL 3)

Crop: Oats, silage-soft dough — Plant date: 10/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	76.79	14.02	62.94	410.97	2,592,000.00 *gal*
Application event totals		76.79	14.02	62.94	410.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	76.79	14.02	62.94	410.97	2,592,000.00 *gal*
Application event totals		76.79	14.02	62.94	410.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	76.79	14.02	62.94	410.97	2,592,000.00 *gal*
Application event totals		76.79	14.02	62.94	410.97	

#1 (VL 3) - 06/08/2023: Corn, silage

Field name: #1 (VL 3)

Crop: Corn, silage — Plant date: 06/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.69	7.96	432.60	432.60	2,880,000.00 *gal*
Ag Well Central	Ground water	29.80	0.00	0.00	1,281.92	19,950,000.00 *gal*
Application event totals		68.49	7.96	432.60	1,714.52	

#1 (VL 3) - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Central	Ground water	29.80	0.00	0.00	1,281.92	19,950,000.00 *gal*
Application event totals		29.80	0.00	0.00	1,281.92	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	42.56	8.75	475.87	475.87	3,168,000.00 *gal*
Ag Well Central	Ground water	24.52	0.00	0.00	1,054.83	16,416,000.00 *gal*
Application event totals		67.08	8.75	475.87	1,530.70	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.69	7.96	432.60	432.60	2,880,000.00 *gal*
Ag Well Central	Ground water	20.43	0.00	0.00	879.03	13,680,000.00 *gal*
Application event totals		59.13	7.96	432.60	1,311.63	

08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	30.96	6.36	346.08	346.08	2,304,000.00 *gal*
Ag Well Central	Ground water	20.43	0.00	0.00	879.03	13,680,000.00 *gal*
Application event totals		51.39	6.36	346.08	1,225.11	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	30.96	6.36	346.08	346.08	2,304,000.00 *gal*
Ag Well Central	Ground water	18.39	0.00	0.00	791.13	12,312,000.00 *gal*
Application event totals		49.35	6.36	346.08	1,137.21	

#1 (VL 3) - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Central	Ground water	18.39	0.00	0.00	791.13	12,312,000.00 *gal*
Application event totals		18.39	0.00	0.00	791.13	

#2 (VL 1) - 10/12/2022: Oats, silage-soft dough

Field name: #2 (VL 1)

Crop: Oats, silage-soft dough

Plant date: 10/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.69	7.96	432.60	432.60	1,728,000.00 *gal*
Application event totals		38.69	7.96	432.60	432.60	

11/07/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	85.32	15.57	69.94	456.64	1,728,000.00 *gal*
Application event totals		85.32	15.57	69.94	456.64	

01/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	85.32	15.57	69.94	456.64	1,728,000.00 *gal*
Application event totals		85.32	15.57	69.94	456.64	

#2 (VL 1) - 06/09/2023: Corn, silage

Field name: #2 (VL 1)

Crop: Corn, silage

Plant date: 06/09/2023

#2 (VL 1) - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	58.04	11.93	648.91	648.91	2,592,000.00 *gal*
Canal Water	Surface water	0.00	0.00	0.00	366.76	7,755,802.57 *gal*
Application event totals		58.04	11.93	648.91	1,015.67	

06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	58.04	11.93	648.91	648.91	2,592,000.00 *gal*
Canal Water	Surface water	0.00	0.00	0.00	366.76	7,755,802.57 *gal*
Application event totals		58.04	11.93	648.91	1,015.67	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	366.76	7,755,802.57 *gal*
Application event totals		0.00	0.00	0.00	366.76	

07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	51.59	10.61	576.81	576.81	2,304,000.00 *gal*
Canal Water	Surface water	0.00	0.00	0.00	275.07	5,816,851.92 *gal*
Application event totals		51.59	10.61	576.81	851.88	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	275.07	5,816,851.92 *gal*
Application event totals		0.00	0.00	0.00	275.07	

#2 (VL 1) - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	51.59	10.61	576.81	576.81	2,304,000.00 *gal*
Canal Water	Surface water	0.00	0.00	0.00	275.07	5,816,851.92 *gal*
Application event totals		51.59	10.61	576.81	851.88	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	51.59	10.61	576.81	576.81	2,304,000.00 *gal*
Canal Water	Surface water	0.00	0.00	0.00	252.15	5,332,114.26 *gal*
Application event totals		51.59	10.61	576.81	828.95	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	252.15	5,332,114.26 *gal*
Application event totals		0.00	0.00	0.00	252.15	

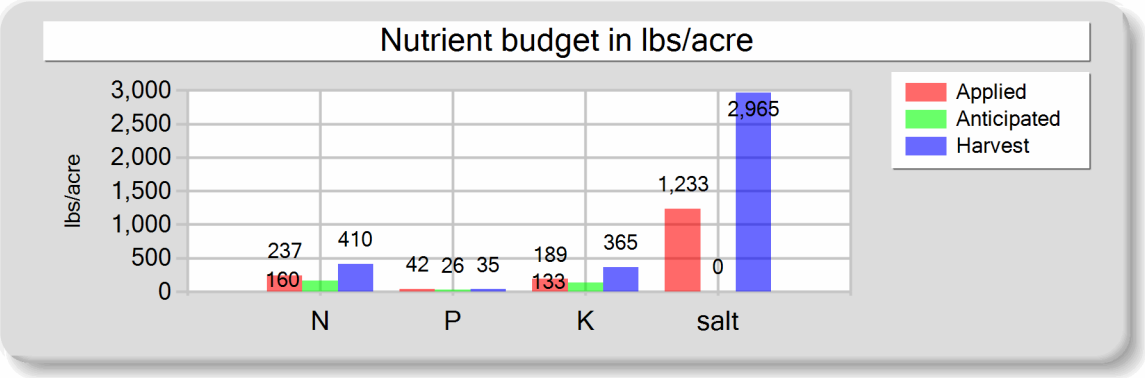
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.00	0.00	0.00	252.15	5,332,114.26 *gal*
Application event totals		0.00	0.00	0.00	252.15	

B. NUTRIENT BUDGET

#1 (VL 3) - 10/10/2022: Oats, silage-soft dough

Field name: #1 (VL 3) Crop: Oats, silage-soft dough Plant date: 10/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	230.36	42.05	188.83	1,232.92
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	237.36	42.05	188.83	1,232.92
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	409.50	35.22	365.27	2,964.78
Nutrient balance	-172.14	6.83	-176.44	-1,731.86
Applied to removed ratio	0.58	1.19	0.52	0.42

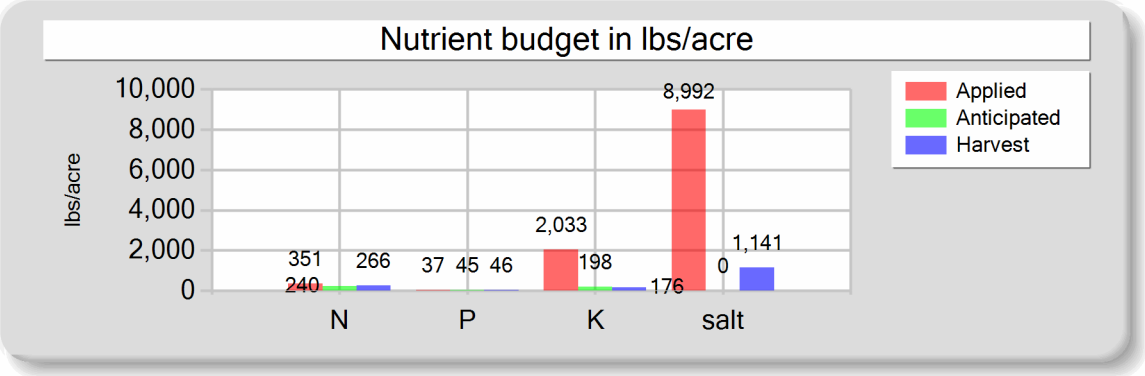
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
7,776,000.00	*gallons*
286.36	*acre-inches*
2.86	*inches/acre*

Total harvests for the crop	
1	*harvests*

#1 (VL 3) - 06/08/2023: Corn, silage

Field name: #1 (VL 3) Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	181.86	37.39	2,033.24	2,033.24
Fresh water	161.77	0.00	0.00	6,958.98
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	350.64	37.39	2,033.24	8,992.22
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	265.61	45.81	176.11	1,141.36
Nutrient balance	85.02	-8.42	1,857.13	7,850.86
Applied to removed ratio	1.32	0.82	11.55	7.88

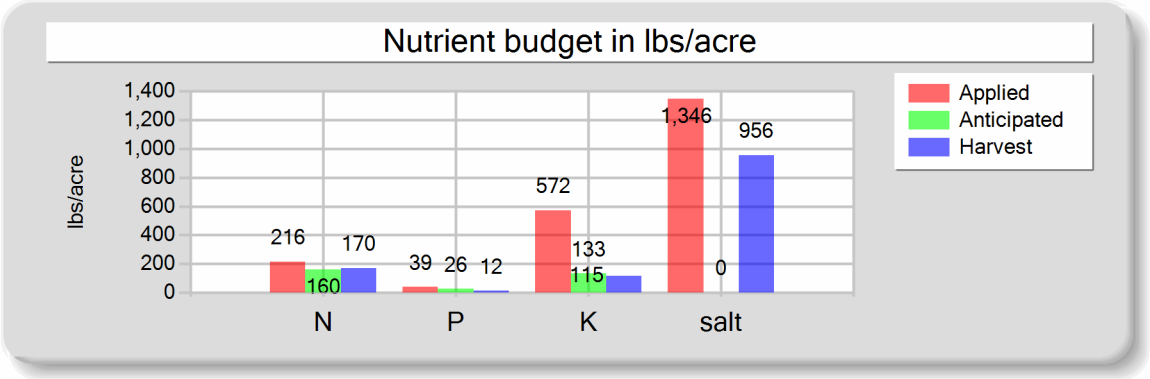
Fresh water applied	
108,300,000.00	*gallons*
3,988.32	*acre-inches*
39.88	*inches/acre*

Process wastewater applied	
13,536,000.00	*gallons*
498.48	*acre-inches*
4.98	*inches/acre*

Total harvests for the crop	
1	*harvests*

#2 (VL 1) - 10/12/2022: Oats, silage-soft dough

Field name: #2 (VL 1) Crop: Oats, silage-soft dough Plant date: 10/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	209.33	39.10	572.48	1,345.88
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	216.33	39.10	572.48	1,345.88
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	169.56	11.99	115.02	955.80
Nutrient balance	46.77	27.11	457.46	390.08
Applied to removed ratio	1.28	3.26	4.98	1.41

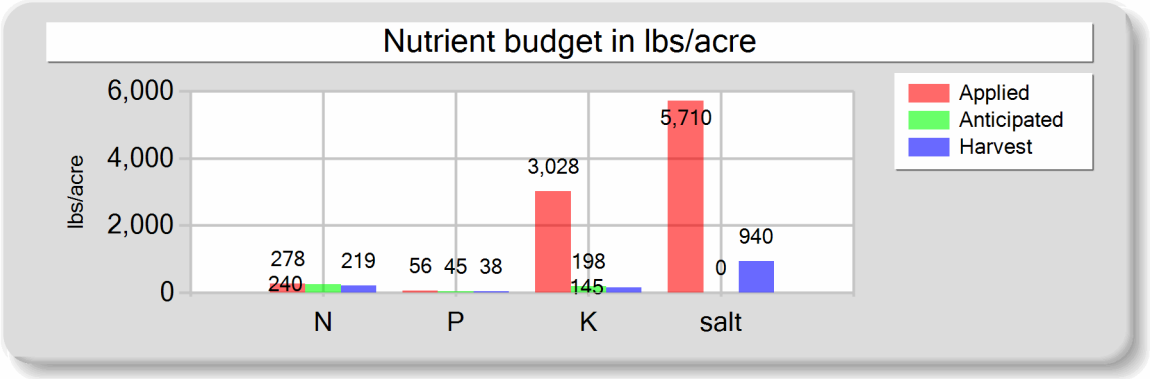
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
5,184,000.00 *gallons*
190.91 *acre-inches*
3.18 *inches/acre*

Total harvests for the crop
1 *harvests*

#2 (VL 1) - 06/09/2023: Corn, silage

Field name: #2 (VL 1) Crop: Corn, silage Plant date: 06/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	270.86	55.69	3,028.23	3,028.23
Fresh water	0.00	0.00	0.00	2,681.93
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	277.86	55.69	3,028.23	5,710.16
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	218.80	37.74	145.07	940.21
Nutrient balance	59.06	17.95	2,883.16	4,769.95
Applied to removed ratio	1.27	1.48	20.87	6.07

Fresh water applied	
56,714,306.25	*gallons*
2,088.60	*acre-inches*
34.81	*inches/acre*

Process wastewater applied	
12,096,000.00	*gallons*
445.45	*acre-inches*
7.42	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

No manure analyses entered.

B. PROCESS WASTEWATER ANALYSES

Lagoon Water 2302052-01

Sample and source description: Lagoon Water 2302052-01

Sample date: 01/31/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	355.00	355.00			64.80	291.00								3,310.00	1,900
DL	0.60	0.60			0.10	10.00								1.00	1

Lagoon Water 2303399-01

Sample and source description: Lagoon Water 2303399-01

Sample date: 03/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	401.00	401.00			72.60	430.00								3,340.00	2,500
DL	0.60	5.00			0.10	10.00								1.00	1

Lagoon Water 2307236-01

Sample and source description: Lagoon Water 2307236-01

Sample date: 07/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	161.00	160.00			33.10	1,800.00								2,990.00	1,800
DL	0.60	0.60			0.10	1.00								1.00	1

Lagoon Water 2310019-01

Sample and source description: Lagoon Water 2310019-01

Sample date: 09/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	366.00	266.00			83.10	478.00								4,300.00	2,700
DL	0.60	0.60			0.10	10.00								1.00	1

C. FRESH WATER ANALYSES

Ag Well Central

2312511-05

Sample description: 2312511-05

Sample date: 12/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	17.90										1,180.00	770
DL	1.40										1.00	1

Ag Well New Well

2312511-03

Sample description: 2312511-03

Sample date: 12/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.87										528.00	320
DL	1.40										1.00	1

Ag Well North

Ag Well North

2312511-04

Sample description: 2312511-04

Sample date: 12/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	7.05										661.00	400
DL	1.40										1.00	1

Canal Water

2102816-03

Sample description: 2102816-03

Sample date: 10/28/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value											453.00	340
DL											1.00	1

Domestic Well Barn

2312511-01

Sample description: 2312511-01

Sample date: 12/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			2.89								592.00	
DL			0.40								1.00	

Domestic Well House J's

Domestic Well House J's

2312511-02

Sample description: 2312511-02

Sample date: 12/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			18.80								1,040.00	
DL			0.40								1.00	

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

#1 (VL 3) - 10/10/2022: Oats, silage-soft dough

2305242-02

Sample and source description: 2305242-02

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 18.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,000.00	2,150.00	22,300.00		18.10
DL	1.00	0.50	0.50		0.10

#1 (VL 3) - 06/08/2023: Corn, silage

#1 (VL 3) - 06/08/2023: Corn, silage

2310040-02

Sample and source description: 2310040-02

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,800.00	2,380.00	9,150.00		5.93
DL	0.50	0.50	0.50		0.10

#2 (VL 1) - 10/12/2022: Oats, silage-soft dough

2305242-01

Sample and source description: 2305242-01

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	31,400.00	2,220.00	21,300.00		17.70
DL	0.50	0.50	0.50		0.10

#2 (VL 1) - 06/09/2023: Corn, silage

2310040-01

Sample and source description: 2310040-01

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,700.00	2,080.00	11,600.00		6.84
DL	0.50	0.50	0.50		0.10

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

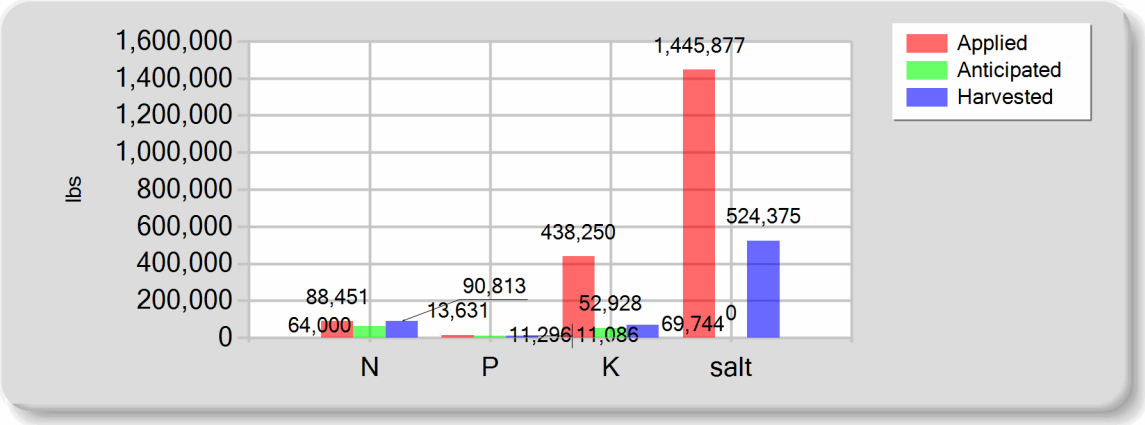
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

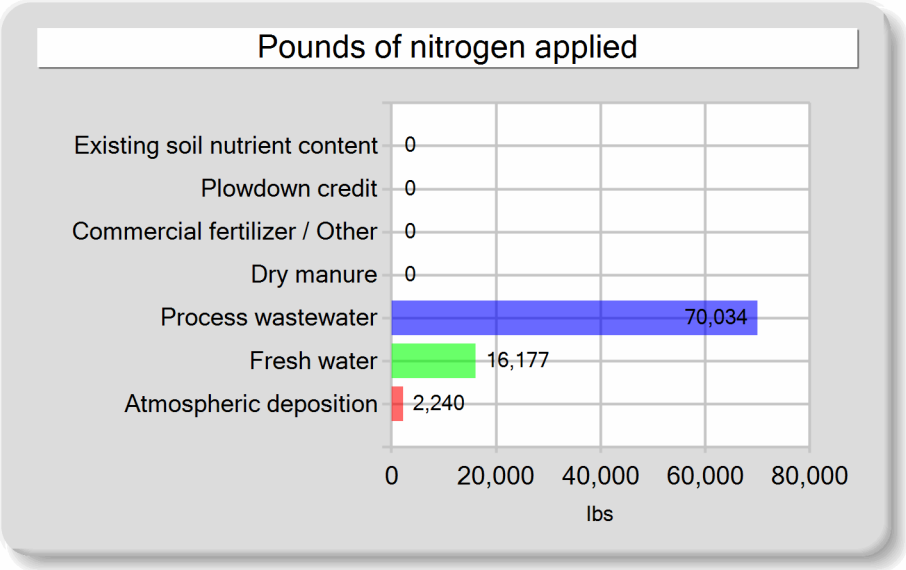
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

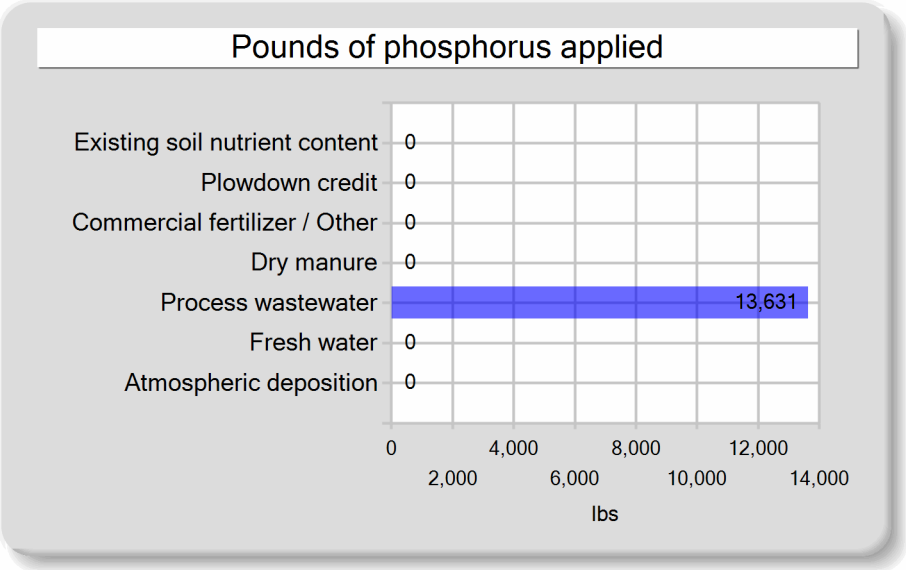
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	70,033.91	13,631.14	438,250.29	589,063.54
Fresh water	16,177.37	0.00	0.00	856,813.40
Atmospheric deposition	2,240.00	0.00	0.00	0.00
Total nutrients applied	88,451.28	13,631.14	438,250.29	1,445,876.93
Anticipated crop nutrient removal	64,000.00	11,296.00	52,928.00	0.00
Actual crop nutrient removal	90,812.82	11,085.93	69,744.28	524,374.60
Nutrient balance	-2,361.54	2,545.20	368,506.01	921,502.33
Applied to removed ratio	0.97	1.23	6.28	2.76

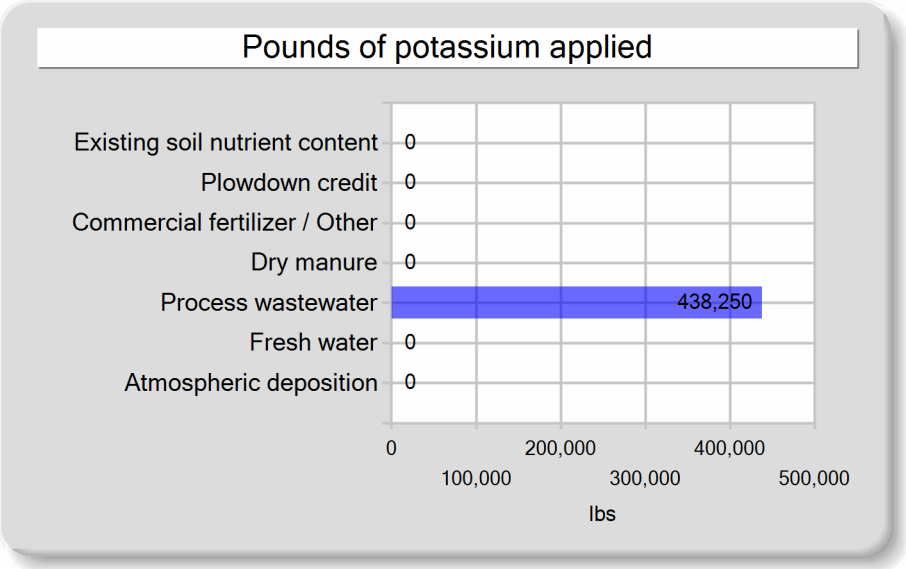
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

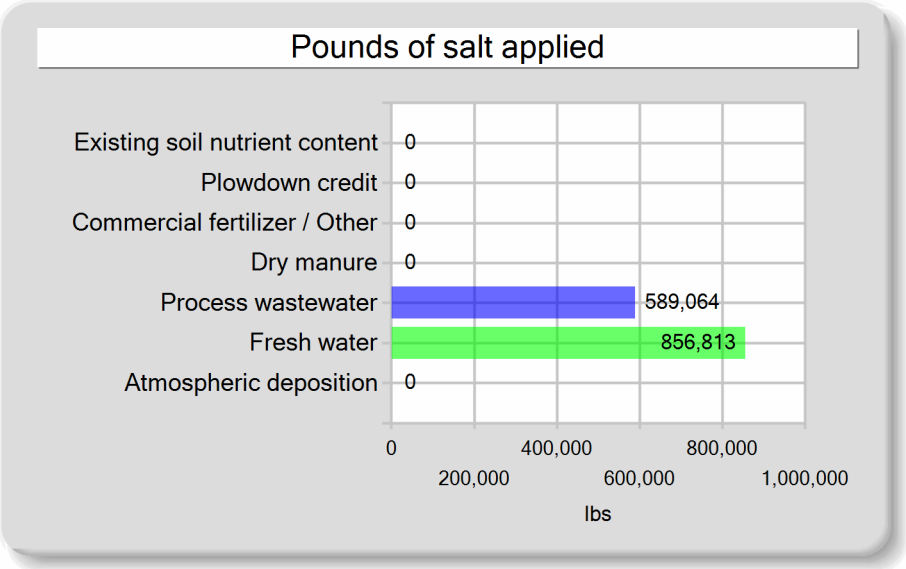


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

X	X
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Jeanette Veldhuis	Paul Van Waremerdam
PRINT OR TYPE NAME	PRINT OR TYPE NAME
✓ Jeanette Veldhuis	✓ 6/27/24
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment	N/A
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .	

Manure/Process Wastewater Tracking Manifests	N/A
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.	

Corrective Actions Documents	N/A
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.	

Groundwater Monitoring	N/A
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.	

Storm Water Monitoring	N/A
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.	